C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
`http://c-immsim.iac.rm.cnr.it (alias to http://kraken.iac.rm.cnr.it/C-IMMSIM)`

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
tt doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
`https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262i`

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
`https://jitc.bmj.com/content/9/5/e002694`

Original C-IMMSIM model: `www.iac.cnr.it/∼filippo/c-immsim`

GETTING HELP:
Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)
Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

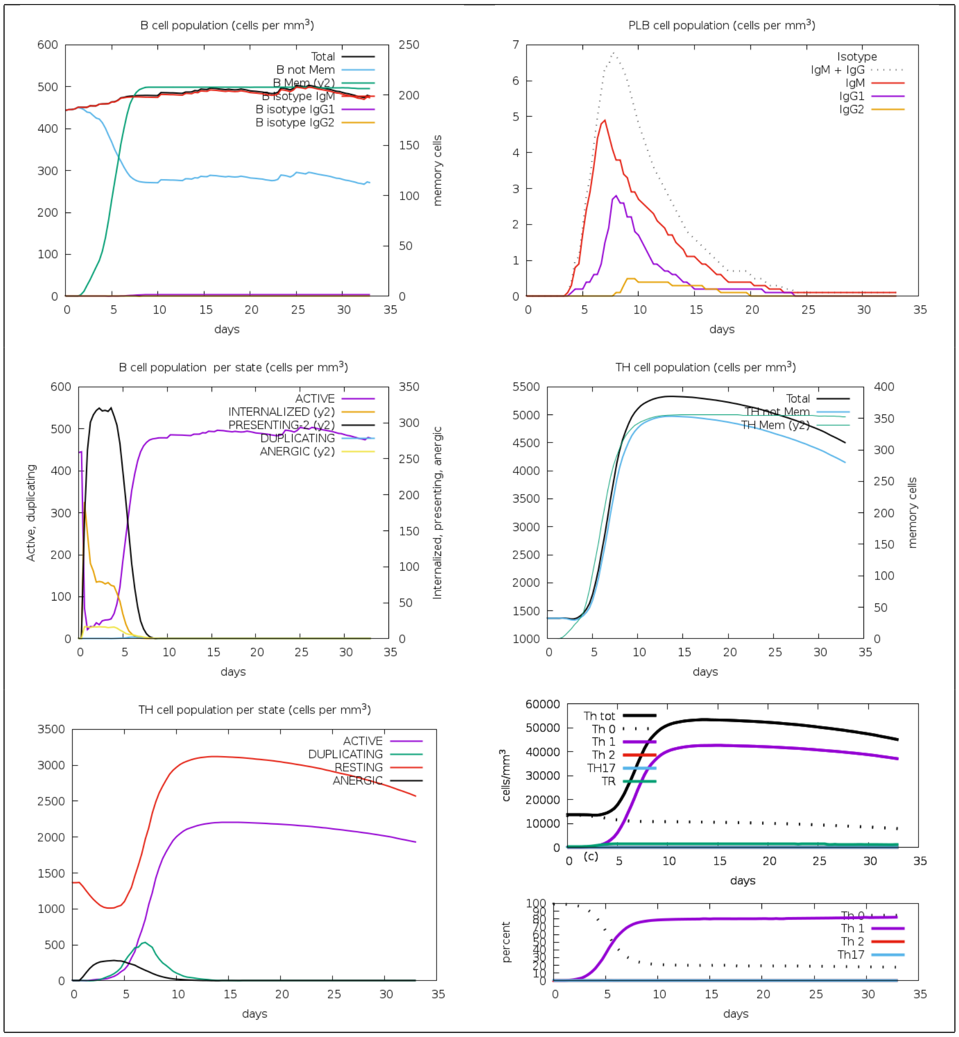

Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

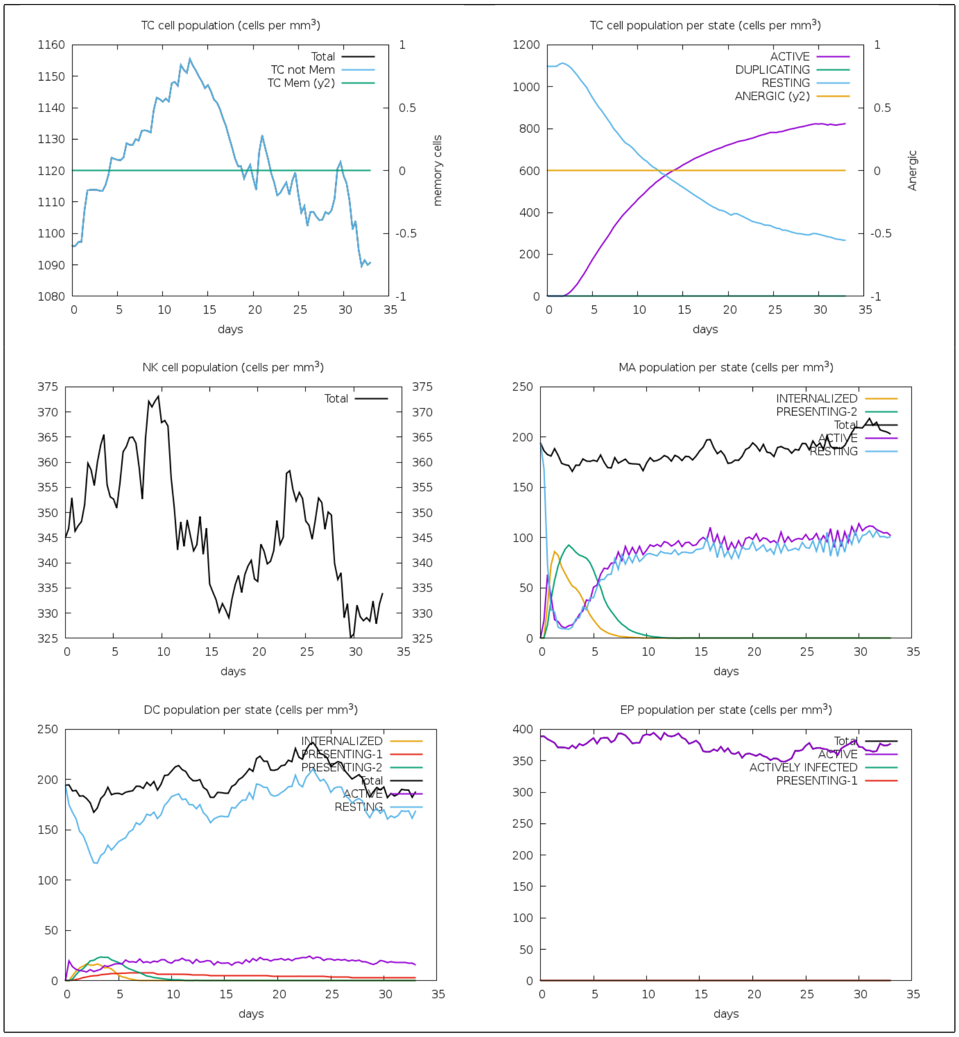

Figure 2: Legend: symbols as figure above.

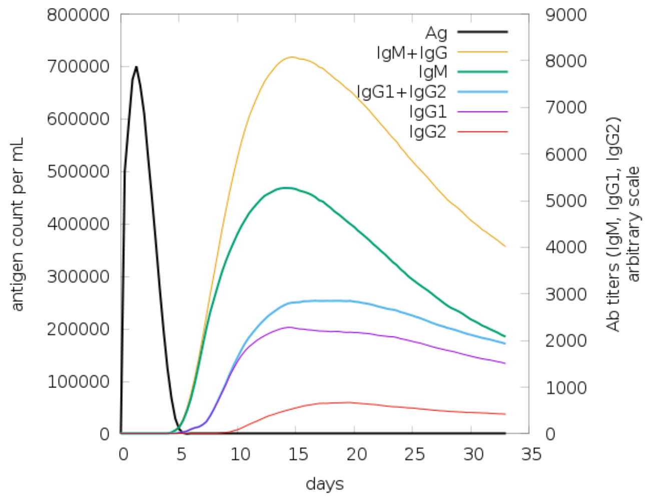

Figure 3: The virus, the immunoglobulins and the immunocomplexes.

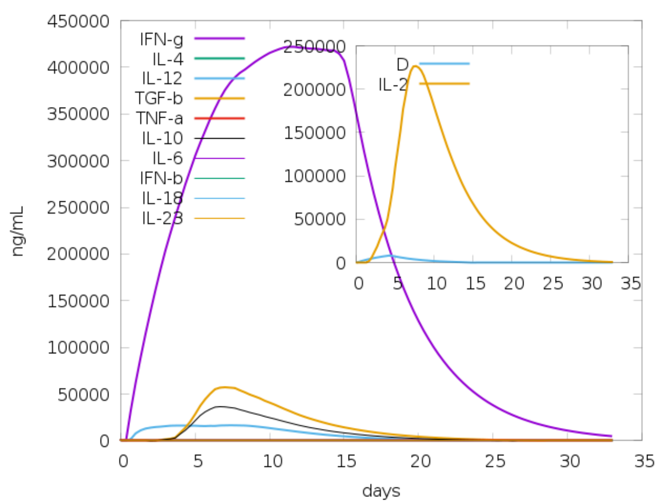

Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

```
Parker_B_scale_epitope_find
Use Parker's propensity scale, takes an antigen block as input,
and creates a list of residues that are possible epitopes.


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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015620_5_xwkdezxgOkHw.FSA_1_001
----------------------------------


APPHALSGGSPEPLPQGQLTEAAAKGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQPAPENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAAAKPPM

00000000010000000000001001111110001111111111111111111111111111111111111111111111
11111110111100111000000011111111111111111111111111111111111111111111111111111111
11111111110000000000000000000000000000000000000000000001000000000000000000000000
0000000110000000000110000000000000000001111111111111111111111000000000
Epitopes of protein 0 ------------
1]       pos=25 len=6       GPGTGP
2]       pos=34 len=53       LGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRH
3]       pos=88 len=4       DGVR
4]       pos=104 len=66       GTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAG
5]       pos=279 len=22       TQNQQTNQAGGEAPQPGDNSTV
```

```
DoPeptideList_I:
        Given the antigen injected creates the list of peptides for all the
        NumAgProts proteins and for all i.e., 4 MHCI molecules

        Read class I peptide list from file? NO

=========================================================================

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000


----------------------------------
Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015620_5_xwkdezxgOkHw.FSA_1_001
----------------------------------

APPHALSGGSPEPLPQGQLTEAAAKGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQPAPENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 0 ------------
0]       pos=  -1 score=1.000000 unnormalised=104.8410000000       non-binding event

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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015620_5_xwkdezxgOkHw.FSA_1_001
----------------------------------

APPHALSGGSPEPLPQGQLTEAAAKGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQPAPENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 0 ------------
0]       pos=  -1 score=1.000000 unnormalised=104.8410000000        non-binding event

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Allele: B0702
Pseudo sequence: KAAREEQQIKAQTRE
Threshold: 8.702800
Max score: 28.406000


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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015620_5_xwkdezxgOkHw.FSA_1_001
----------------------------------

APPHALSGGSPEPLPQGQLTEAAAKGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQPAPENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 0 ------------
0]       pos=  13 score=0.005862 unnormalised=0.6622000000        LPQGQLTEA
1]       pos=  73 score=0.002667 unnormalised=0.3012000000        RPGAPGGSG
2]       pos=  92 score=0.017106 unnormalised=1.9322000000        RPQKRPSCI
3]       pos= 216 score=0.013875 unnormalised=1.5672000000        GPGLPQGQL
4]       pos= 247 score=0.021019 unnormalised=2.3742000000        APTGSWFSA
5]       pos= 255 score=0.002649 unnormalised=0.2992000000        APQPAPENA
6]       pos= 291 score=0.008660 unnormalised=0.9782000000        APQPGDNST
7]       pos=  -1 score=0.928163 unnormalised=104.8410000000        non-binding event

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Allele: B0702
Pseudo sequence: KAAREEQQIKAQTRE
Threshold: 8.702800
Max score: 28.406000


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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015620_5_xwkdezxgOkHw.FSA_1_001
----------------------------------

APPHALSGGSPEPLPQGQLTEAAAKGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQPAPENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 0 ------------
0]       pos=  13 score=0.005862 unnormalised=0.6622000000        LPQGQLTEA
1]       pos=  73 score=0.002667 unnormalised=0.3012000000        RPGAPGGSG
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3]       pos= 216 score=0.013875 unnormalised=1.5672000000        GPGLPQGQL
4]       pos= 247 score=0.021019 unnormalised=2.3742000000        APTGSWFSA
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```

```
6]        pos= 291 score=0.008660 unnormalised=0.9782000000        APQPGDNST
7]        pos=  -1 score=0.928163 unnormalised=104.8410000000        non-binding event
```

```
DoPeptideList_II:
        Given the antigen injected creates the list of peptides for all the
        NumAgProts proteins and for all i.e., 2 MHCII molecules

        Read class II peptide list from file? NO

========================================================================

Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000


----------------------------------
Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015620_5_xwkdezxgOkHw.FSA_1_001
----------------------------------

APPHALSGGSPEPLPQGQLTEAAAKGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQPAPENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 0 -----------
0]        pos=   5 score=0.035276 unnormalised=3.1995600000        LSGGSPEPL
1]        pos= 181 score=0.018452 unnormalised=1.6735600000        VVLLGPGPG
2]        pos= 182 score=0.009389 unnormalised=0.8515600000        VLLGPGPGT
3]        pos= 198 score=0.032300 unnormalised=2.9295600000        FGPGPGACM
4]        pos= 211 score=0.009830 unnormalised=0.8915600000        LEKGPGPGL
5]        pos= 239 score=0.069191 unnormalised=6.2755600000        YTLGPGPGA
6]        pos= 241 score=0.002608 unnormalised=0.2365600000        LGPGPGAPT
7]        pos= 264 score=0.093425 unnormalised=8.4735600000        YQAYAAPQL
8]        pos=  -1 score=0.729530 unnormalised=66.1680000000        non-binding event

========================================================================

Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000


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----------------------------------

APPHALSGGSPEPLPQGQLTEAAAKGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQPAPENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 0 -----------
0]        pos=   5 score=0.035276 unnormalised=3.1995600000        LSGGSPEPL
1]        pos= 181 score=0.018452 unnormalised=1.6735600000        VVLLGPGPG
2]        pos= 182 score=0.009389 unnormalised=0.8515600000        VLLGPGPGT
3]        pos= 198 score=0.032300 unnormalised=2.9295600000        FGPGPGACM
4]        pos= 211 score=0.009830 unnormalised=0.8915600000        LEKGPGPGL
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5]        pos= 239 score=0.069191 unnormalised=6.2755600000        YTLGPGPGA
6]        pos= 241 score=0.002608 unnormalised=0.2365600000        LGPGPGAPT
7]        pos= 264 score=0.093425 unnormalised=8.4735600000        YQAYAAPQL
8]        pos=  -1 score=0.729530 unnormalised=66.1680000000        non-binding event
```
